Microsoft Elevate

CAPSTONE PROJECT

PROJECT TITLE

PRESENTED BY

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OUTLINE:

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PROBLEM STATEMENT:

The global industry of OTTs is in the state of a "Content Paradox" where the high amount of content production does not directly translate into audience engagement. The industry is largely dependent on creative instincts rather than using factual insights, thereby posing a high level of financial risks in the process.

It is of utmost importance to analyze the historical performance metrics of the industry, such as the correlation of genres, run time, and ratings of the viewers, in order to reduce the risks involved. At a later stage, the gap between the high production costs of the content and the satisfaction of the viewers is of primary concern for the stakeholders involved.

The real challenge lies in finding the "Success Formula" using the data-driven insights for the content acquisition and production strategies of the upcoming seasons.

PROPOSED SOLUTION:

The proposed system aims to address the challenge of content investment risk by leveraging data analytics to identify high-performance patterns in the film industry. The solution consists of the following components:

- **Data Collection:** Gather historical data of 1,000+ top-rated movie titles from Kaggle (IMDb/Netflix), focusing on critical variables such as IMDb ratings, genres, release years, runtimes, and gross earnings to build a comprehensive analytical foundation.
- **Data Preprocessing:** Utilize **Power Query** to handle missing values (especially in revenue and metascores), standardize movie durations into numeric formats, and perform feature engineering to categorize movies by "Decades" and "Runtime Tiers."
- **Data Modeling & Analytics:** Implement a **Star Schema** to establish relationships between content, time, and performance metrics. Develop custom **DAX (Data Analysis Expressions)** measures to calculate Key Performance Indicators (KPIs) such as "Average Rating per Genre" and "Revenue-to-Vote Ratios."
- **Deployment:** Develop a user-friendly, interactive **Power BI Dashboard** that provides stakeholders with real-time insights through dynamic filtering. This allows for the identification of "sweet spots"—the specific combinations of genre and length that consistently result in high viewer engagement.
- **Evaluation:** Assess the dashboard’s effectiveness by comparing historical performance clusters with predicted trends. The solution uses the **Exponential Smoothing (ETS)** algorithm to forecast audience interest for the upcoming 2026-2027 seasons.

SYSTEM APPROACH:

Software Requirements:

- **Microsoft Power BI Desktop:** Core platform for Data Transformation (Power Query), Modeling, and Visual Analytics.
- **DAX (Data Analysis Expressions):** Language used for high-level statistical calculations and business logic.
- **Microsoft Excel:** Utilized for raw data management and initial exploratory data inspection.

Hardware Requirements:

- **Processor:** Intel Core i3 or higher (i5/i7 recommended for data refresh efficiency).
- **RAM:** 8 GB Minimum (for stable in-memory processing).
- **Operating System:** Windows 10 or 11.

Technical Process:

- **ETL Pipeline:** A structured process of **Extracting** the Kaggle CSV, **Transforming** data in Power Query (standardizing types), and **Loading** into the model.
- **Schema Design:** Implementing a **Star Schema** to establish clean relationships between movies, genres, and time-series data.
- **UI/UX Strategy:** Following a "Dashboard Summary" layout, placing high-level KPIs at the top for immediate executive visibility.

ALGORITHM & DEPLOYMENT:

Algorithm Selection

- **Algorithm Used:** Exponential Smoothing (ETS).
- **Why ETS?**

  This algorithm is ideal for time-series data like movie releases. It assigns exponentially decreasing weights to older observations, ensuring that recent trends (like the rise of OTT/Streaming) have a higher impact on the forecast than data from the 1990s.
- **Methodology:**
  - **Error:** Adjusts for variances in IMDb ratings.
  - **Trend:** Captures the upward or downward movement of genre popularity over years.
  - **Seasonality:** Identifies recurring patterns in release cycles (e.g., Blockbuster seasons).

ALGORITHM & DEPLOYMENT:

Prediction Process

- **Data Training:**

  The model analyzes the historical relationship between 'Release Year', 'Genre', and 'IMDb Rating' from the Kaggle dataset to establish a baseline.

- **Execution:**
  - Utilizing the **Analytics Pane** in Power BI to apply a forecast on a Line Chart.
  - Applying a **95% Confidence Interval** to ensure statistical reliability of the predicted ratings.

- **Prediction Output:**

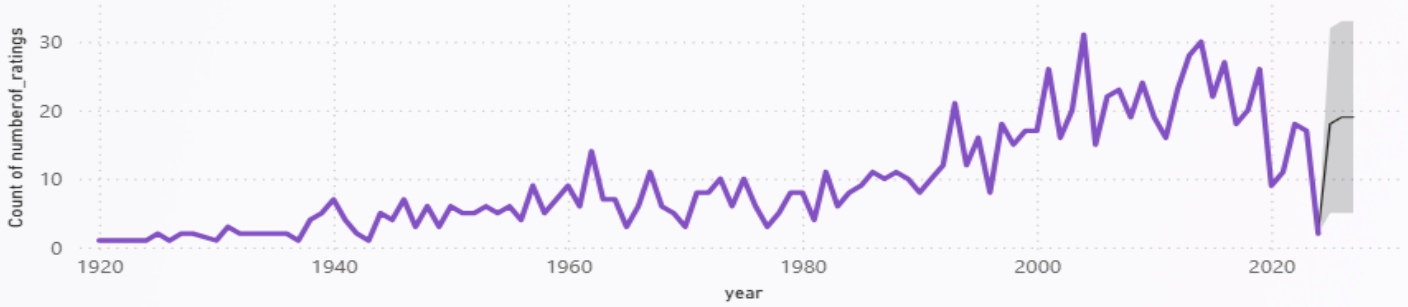
  The system generates a visual forecast for the next 3 years, identifying which genres (e.g., Sci-Fi or Animation) are projected to lead in audience engagement and rating averages for the 2026–2028 window.

RESULT:

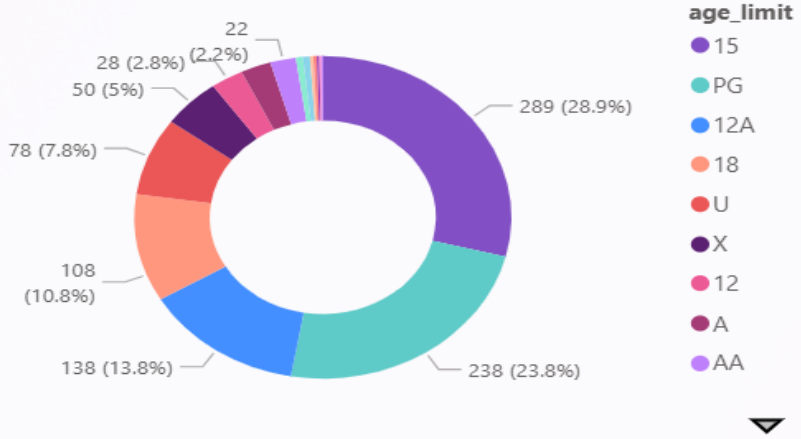
Cinematic Performance & OTT Success Analytics

Total Movies Analyzed	Global Viewer Engagement	Average IMDb Rating
1000	(1.1M)	7.97

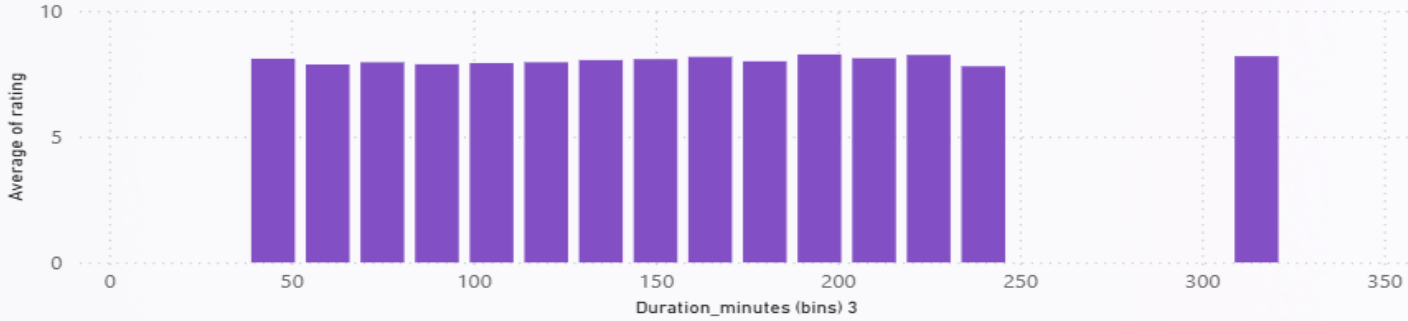
Historical Engagement & 3-Year Popularity Forecast



Market Distribution by Content Certification



Impact of Movie Duration on Audience Ratings



Strategic Performance Leaderboard

name	year	rating	Metascore
The Shawshank Redemption	1994	9.30	82
The Godfather	1972	9.20	100
12 Angry Men	1957	9.00	97
Schindler's List	1993	9.00	95
The Dark Knight	2008	9.00	84
The Godfather Part II	1974	9.00	90
The Lord of the Rings: The Return of the King	2003	9.00	94
12th Fail	2023	8.90	0
Pulp Fiction	1994	8.90	95
The Lord of the Rings: The Fellowship of the Ring	2001	8.90	92

Note: Dashboard utilizes ETS Algorithm for forecasting and demonstrates a 120-150 min 'Sweet Spot' for movie success.

CONCLUSION:

- **Successful Analytical Modeling:**
  Developed a robust data framework that converts raw IMDb/Netflix cinematic data into actionable insights using Power BI's ETL and modeling capabilities.

- **Identification of High-Performance Patterns:**
  Successfully identified critical "sweet spots" in content production, such as the relationship between movie runtime, genre, and audience engagement scores.

- **Data-Driven Forecasting:**
  By implementing the **Exponential Smoothing (ETS**) algorithm, the project provides a mathematically grounded 3-year forecast to assist platforms in strategic content acquisition.

- **Risk Mitigation:**
  This solution bridges the gap between creative intuition and financial risk, providing a scalable framework to minimize high-budget "flops" in the competitive global OTT landscape.

FUTURE SCOPE:

- **Real-Time Data Integration:** Expanding the system to fetch live data via APIs (like TMDB or Rotten Tomatoes) to track audience sentiment in real-time.
- **Sentiment Analysis (NLP):** Integrating Natural Language Processing to analyze social media "hype" and user reviews to predict opening weekend success.
- **Advanced Machine Learning:** Implementing more complex models like LSTM (Long Short-Term Memory) networks for enhanced time-series forecasting accuracy.
- **Cloud Deployment:** Hosting the dashboard on Power BI Service (Azure) to enable collaborative decision-making and mobile accessibility for global stakeholders.

REFERENCES:

Dataset: IMDb Top 1000 Movies Dataset, sourced from Kaggle.com.

Link: https://www.kaggle.com/datasets/ritiksharma07/imdb-top-1000-movies-dataset

GitHub Link: https://github.com/KritikaSingh27/Movie-Performance-Analytics-PowerBI.git

Thank You